

Smart Timer For LIQUID MOSQUITO REPELLENTS

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As the hazards of mosquitoes are increasing day by day, the usage of liquid (electric) mosquito repellents is on the rise as well. These devices are widely employed for ensuring a good night's sleep and are also commonly found in offices to enhance work environments. Most liquid mosquito repellents are ruggedly constructed and operate continuously once activated. But the continuous operation of these mosquito repellents causes the vapour content in the room to exponentially increase over time, which can cause harmful breathing effects and allergic reactions.

The smart timing operation for liquid mosquito repellents addresses this issue, offering different modes suitable for residential and office purposes. The author's prototype is shown in Fig. 1, with the required components listed in Table 1.

Circuit and working

The circuit diagram of the smart timer for liquid mosquito repellents is shown in Fig. 2. It is built around the Arduino Uno board (MOD1), RTC DS1307 module (MOD2), 5V relay module, 5V buzzer (B1), and a few other components. The positive terminal of buzzer (B1) is connected to pin D13 of the Arduino Uno, with its ground pin connected to the Arduino ground. A 12V DC adaptor that powers the circuit is directly connected to the DC socket on the Arduino Uno board.

The relay's normally-open (NO) pins are connected to a 220V AC, 6A electrical socket, which is also

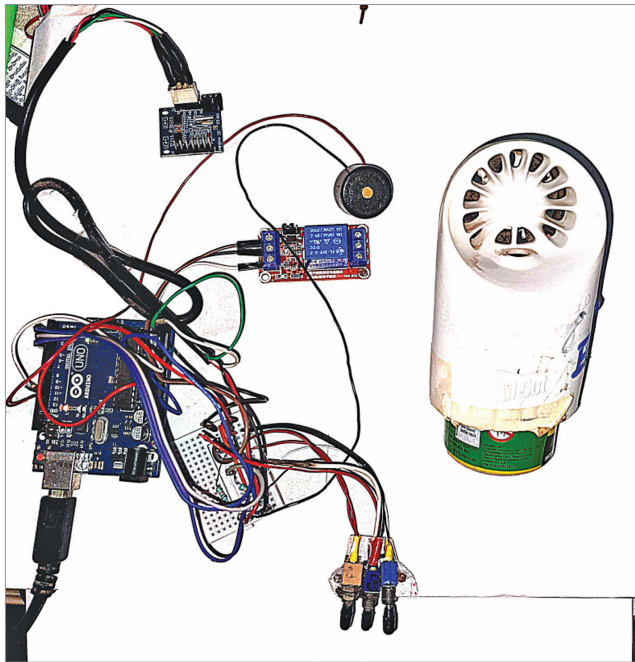


Fig. 1: Author's prototype

TABLE 1
BILL OF MATERIALS

Components	Quantity
Arduino Uno board (MOD1)	1
DS1307 RTC module (MOD2)	1
One channel 5V relay module	1
10-kilo-ohm resistor (R1, R2, R3)	3
On/off switch (S1 through S4)	4
12V DC adaptor	1

used to connect the liquid mosquito repellents. An additional 6A SPST electrical switch (S4) is connected across the NO and COM pins of the relay module, which acts as a bypass switch whenever smart switching action is not needed. Images of the DS1307 I2C RTC module and the 5V relay module are given

in Fig. 3 and Fig. 4, respectively.

The Arduino board, which is the heart of the system, is central to the real-time clock, mode selection switches, and the 5V relay board. The main purpose of the timer is to regulate the on/off cycles of the liquid mosquito repellent according to different modes of operation. For convenience, it has seven modes that can be selected

by three switches.

Mode 1: Night mode. In this general-purpose mode, the controller automatically switches on the mosquito repellent from 6:00pm to 6:00am for uninterrupted sleep. To enable this mode, keep switches S1, S2, and S3 open.

Mode 2: Night toggle mode. In this mode, the timer switches on the repellent continuously from 6:00pm to 00:00 hours, then toggles from 00:00 hours to 6:00pm, every hour. To enable this mode, close switch S1 and keep switches S2 and S3 open.

Mode 3: Night toggle mode with buzzer. It is similar to mode 2, but has an added feature of a buzzer acting as a wake-up alarm at 6:00am. To enable this mode, close switch S2